

Algebra Exercises

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Chapter 6

Linear Algebra

6.1 Free Modules Revisited

Exercise 6.1.1. Show $\mathbb{R} \cong \mathbb{C}$ as $\mathbb{Q}$ -vector spaces.

Solution. This is tricky. I think $\dim_{\mathbb{Q}} \mathbb{R} = \aleph_0$, and $\dim_{\mathbb{Q}} \mathbb{C} = 2 \dim_{\mathbb{Q}} \mathbb{R} = \dim_{\mathbb{Q}} \mathbb{R}$. Unsure how to show this. Perhaps we can note that

$$\mathbb{C} \cong \frac{\mathbb{R}[x]}{(x^2 + 1)},$$

and use this fact somehow. Or more notably, we can use the fact that $\mathbb{C} \cong \mathbb{R} \oplus \mathbb{R}$ as groups. Waiiiiit recall that

$$F^{\mathbb{Q}}(\mathbb{R}) \cong F^{\mathbb{Q}}\mathbb{C} \iff \mathbb{R} \cong \mathbb{C}.$$

Ok but these free groups arent isomorphic to $\mathbb{R}, \mathbb{C}$. How do i find a basis? ■

Exercise 6.1.2. Consider the sets

- $\mathfrak{gl}_n(\mathbb{R}) := \mathcal{M}_n(\mathbb{R})$;
- $\mathfrak{sl}_n(\mathbb{R}) := \{M \in \mathfrak{gl}_n(\mathbb{R}) \mid \text{tr } M = 0\}$;
- $\mathfrak{sl}_n(\mathbb{C}) := \{M \in \mathfrak{gl}_n(\mathbb{C}) \mid \text{tr } M = 0\}$;
- $\mathfrak{so}_n(\mathbb{R}) := \{M \in \mathfrak{sl}_n(\mathbb{R}) \mid M + M^t = 0\}$;
- $\mathfrak{su}(n) := \{M \in \mathfrak{sl}_n(\mathbb{C}) \mid M + M^\dagger = 0\}$.

Show these sets are $\mathbb{R}$ -vector spaces, and compute their dimension.

Solution. First, we show $\mathfrak{gl}_n(\mathbb{R})$ is a $\mathbb{R}$ -vector space. We will use the componentwise matrix notation $M := [m_{ij}]$, where m_{ij} are the components of the matrix M and range over $\{1, \dots, n\}$. For any $r \in \mathbb{R}$, define $r[m_{ij}] = [rm_{ij}]$. Clearly, $\mathfrak{gl}_n(\mathbb{R})$ is a group, so we will simply show it is a vector space. Note that $1[m_{ij}] = [m_{ij}]$,

$$(r + s)[m_{ij}] = r[m_{ij}] + s[m_{ij}],$$

and

$$(rs)[m_{ij}] = [rsm_{ij}] = r[sm_{ij}] = r(s[m_{ij}]).$$

Moreover, r is an endomorphism:

$$r([m_{ij}] + [n_{ij}]) = r[m_{ij} + n_{ij}] = [r(m_{ij} + n_{ij})] = [rm_{ij} + rn_{ij}] = [rm_{ij}] + [rn_{ij}] = r[m_{ij}] + r[n_{ij}].$$

Therefore, $r \mapsto r$ is a homomorphism $\mathbb{R} \rightarrow \text{End}_{\text{Ab}}(\mathfrak{gl}_n(\mathbb{R}))$, and $\mathfrak{gl}_n(\mathbb{R})$ is an $\mathbb{R}$ -vector space. Finally, we compute the dimension. Obviously, $\mathfrak{gl}_n(\mathbb{R}) \cong \mathbb{R}^{n^2}$. This is so much effort to prove so I'm not going to.

$\mathfrak{sl}_n(\mathbb{R})$ is very difficult, but note that $\mathfrak{so}_n(\mathbb{R}) \subseteq \mathfrak{sl}_n(\mathbb{R}) \subseteq \mathfrak{gl}_n(\mathbb{R})$, so maybe that helps.

I think the basis for $\mathfrak{so}_n(\mathbb{R})$ is basically one element per item m_{ij} in the upper triangle (not counting main diagonal), put a 1 there, and put a -1 at m_{ji} . Then $M^t = -M$ for the basis set. I think this is the entire basis.

For $\mathfrak{so}_n(\mathbb{R})$, we have $[m_{ij}] = -[m_{ji}]$, and thus the diagonal entries are all 0. It follows that we simply have

$$\dim_{\mathbb{R}} \mathfrak{so}_n(\mathbb{R}) = \frac{n^2 - n}{2}.$$

The n^2 is the number of entries in the matrix, the $-n$ removes the diagonal elements, and the division by 2 reduces it to only one of the remaining two triangles, since $m_{ij} = -m_{ji}$. ■

Exercise 6.1.3. Prove $\mathfrak{su}(2) \cong \mathfrak{so}_3(\mathbb{R})$ as $\mathbb{R}$ -vector spaces. In particular these spaces can be viewed as the tangent spaces to $\text{SU}(2)$ and $\text{SO}_3(\mathbb{R})$, so given the surjective homomorphism $I : \text{SU}(2) \rightarrow \text{SO}_3(\mathbb{R})$ constructed in 2.8.9, there is an induced more meaningful isomorphism.

| **Solution.** h ■

Exercise 6.1.4. Let V be a k -vector space. A *Lie Bracket* on V is an operation $[\cdot, \cdot] : V \times V \rightarrow V$ such that for all $u, v, w \in V$ and $a, b \in k$,

- $[au + bv, w] = a[u, w] + b[v, w]$ and $[w, au + bv] = a[w, u] + b[w, v]$;
- $[v, v] = 0$;
- $[[u, v], w] + [[v, w], u] + [[w, u], v] = 0$.

This last axiom is called the *Jacobi identity*. A vector space endowed with a Lie bracket is called a Lie algebra. Define a category of Lie algebras over a given field. Prove the following:

- In a Lie algebra, $[u, v] = -[v, u]$ for all $u, v \in V$.
- If V is a k -algebra, then $[v, w] := vw - wv$ defines a Lie bracket on V , so that V is a Lie algebra in a natural way.
- This makes $\mathfrak{gl}_n(\mathbb{R})$ and $\mathfrak{gl}_n(\mathbb{C})$ into Lie algebras. In fact, the sets listed in exercise 6.1.2 are all Lie algebras, with respect to a Lie bracket induced from $\mathfrak{gl}$.
- $\mathfrak{su}(2)$ and $\mathfrak{so}_3(\mathbb{R})$ are isomorphic as Lie algebras over $\mathbb{R}$.

Solution. Let $k\text{-Lie}$ be the category of Lie algebras over a field k . Objects in $k\text{-Lie}$ are clearly k -Lie algebras. Morphisms in $k\text{-Lie}$ are k -linear maps $\phi : \mathfrak{u} \rightarrow \mathfrak{v}$ such that

$$\phi([u, v]_{\mathfrak{u}}) = [\phi(u), \phi(v)]_{\mathfrak{v}}.$$

This is a fairly natural request. Note that the identity map satisfies this requirement:

$$1([u, v]) = [u, v] = [1(u), 1(v)].$$

Additionally, for $\phi : \mathfrak{u} \rightarrow \mathfrak{v}$, $\psi : \mathfrak{v} \rightarrow \mathfrak{w}$ morphisms in $k\text{-Lie}$, we have

$$(\psi\phi)([u, v]_{\mathfrak{u}}) = \psi([\phi(u), \phi(v)]_{\mathfrak{v}}) = [(\psi\phi)(u), (\psi\phi)(v)]_{\mathfrak{w}}.$$

Therefore, $\psi\phi : \mathfrak{u} \rightarrow \mathfrak{w}$ is a $k\text{-Lie}$ morphism, and $k\text{-Lie}$ is a category.

Note that in any lie algebra,

$$\begin{aligned} [u, v] + [v, u] &= [u, v] + 0 + [v, u] + 0 \\ &= [u, v] + [u, u] + [v, u] + [v, v] \\ &= [u, v + u] + [v, u + v] \\ &= [u, u + v] + [v, u + v] \\ &= [u + v, u + v] \\ &= 0. \end{aligned}$$

Therefore, $[u, v] = -[v, u]$.

Clearly, the commutator is a Lie bracket:

$$[u, u] = u^2 - u^2 = 0;$$

$$\begin{aligned} [au + bv, w] &= (au + bv)w - w(au + bv) \\ &= auw + bvw - awu - bwv \\ &= auw - awu + bvw - bwv \\ &= a[u, w] + b[v, w]; \end{aligned}$$

$$\begin{aligned} &[[u, v], w] + [[v, w], u] + [[w, u], v] \\ &= (uv - vu)w - w(uv - vu) + (vw - wv)u - u(vw - wv) + (wu - uw)v - v(wu - uw) \\ &= uvw - vuw - wuv + wvu + vwu - wvu - uvw + uwv + wuv - uwv - vwu + vuw \\ &= 0. \end{aligned}$$

I believe it is a Lie bracket in the *natural way* because all $k\text{-Alg}$ morphisms $\phi : V \rightarrow W$ are also Lie algebra morphisms, when V and W are endowed with the commutator:

$$\phi([u, v]) = \phi(uv - vu) = \phi(u)\phi(v) - \phi(v)\phi(u) = [\phi(u), \phi(v)].$$

I think I can skip the next bullet point, since we have shown enough to show these sets are $\mathbb{R}$ -algebras or $\mathbb{C}$ -algebras, and this holds for all k -algebras.

We have already that $\mathfrak{so}_3(\mathbb{R}) \cong \mathfrak{su}(2)$ as $\mathbb{R}$ -vector spaces, so we can simply show they are isomorphic as Lie algebras. Will complete after previous problem. ■

Exercise 6.1.5. Let R be an integral domain. Prove or disprove the following:

- Every linearly independent subset of a free R -module may be completed to a basis.
- Every generating subset of a free R -module contains a basis.

| **Solution.** idk ■

Exercise 6.1.6. Prove that if k is a field and V is a k -vector space, and B is a minimal generating set for V , then B is a basis for V .

| **Solution.** also easy ■

Exercise 6.1.7. Let R be an integral domain, and let $M := R^{\oplus A}$ be a free R -module. Let K be the field of fractions of R , and view M as a subset of $V := K^{\oplus A}$ in the evident way. Prove that a subset $S \subseteq M$ is linearly independent in M over R if and only if it is linearly independent in V over K . Conclude that $\text{rk}_R M = \dim_K V$. Prove that if S generates M over R , then it generates V over K . Is the converse true?

| **Solution.** a ■

Exercise 6.1.8. Recall that for any integral domain R and free R -module M , for any linearly independent subset S and maximally linearly independent subset B of M , $|S| \leq |B|$. In particular, any two maximally linearly independent subsets of a free R -module have the same cardinality. Using this fact, deduce that

$$F^R(A) \cong_{R\text{-Mod}} F^R(B) \iff A \cong_{\text{Set}} B.$$

| **Solution.** idk how to do this ngl ■

Exercise 6.1.9. Let R be a commutative ring, and let M be an R -module. Let $\mathfrak{m}$ be a maximal ideal in R , such that $\mathfrak{m}M = 0$ (that is, $rm = 0$ for all $r \in R$ and $m \in \mathfrak{m}$). Define in a natural way a vector space structure over $R/\mathfrak{m}$ on M .

| **Solution.** ??? ■

Exercise 6.1.10. Let R be a commutative ring, and let $F := R^{\oplus B}$ be a free R -module. Let $\mathfrak{m}$ be a maximal ideal of R , and let $k = R/\mathfrak{m}$ be the quotient field. Prove that $F/\mathfrak{m}F \cong k^{\oplus B}$ as k -vector spaces.

| **Solution.** ??????????????????// ■

Exercise 6.1.11. Show that commutative rings satisfy the IBN property.

| **Solution.** okie ■

Exercise 6.1.12. Let V be a k -vector space, and let $R = \text{End}_{k\text{-Vect}}(V)$.

- Prove that $\text{End}_{k\text{-Vect}}(V \oplus V) \cong R^4$ as an R -module.
- Prove that R does not satisfy the IBN property if $V = k^{\oplus \mathbb{N}}$.

Note that $V \cong V \oplus V$ if $V = k^{\oplus \mathbb{N}}$.

| **Solution.** a ■

Exercise 6.1.13. Let A be an abelian group such that $\text{End}_{\text{Ab}}(A)$ is a field of characteristic 0. Prove that $A \cong \mathbb{Q}$. (Hint: Prove that A carries a $\mathbb{Q}$ -vector space structure, and consider its dimension).

| **Solution.** a ■

Exercise 6.1.14. Let V be a finite-dimensional vector space, and let $\phi : V \rightarrow V$ be a homomorphism of vector spaces. Prove that there exists an integer $n \in \mathbb{Z}$ such that $\ker \phi^{n+1} = \ker \phi$ and $\text{im } \phi^{n+1} = \text{im } \phi^n$. Show that both claims may fail if V has infinite dimension.

Solution. If iso then epic and mono so $n = 1$. If not iso, since it's finite, we can probably do some kind of induction? ■

Exercise 6.1.15. Consider the previous question for free R -modules of finite rank, where R is an integral domain that is not a field. Let $\phi : F \rightarrow F$ be a homomorphism of free R -modules.

- What property immediately guarantees $\ker \phi^{n+1} = \ker \phi^n$ for $n \gg 0$?
- Show there exists an R -module homomorphism $\phi : F \rightarrow F$ such that $\operatorname{im} \phi^{n+1} \subsetneq \operatorname{im} \phi^n$ for all $n \in \mathbb{N}$.

Solution. a ■

Exercise 6.1.16. Let M be a module over a ring R . A *finite composition series* for M is a decreasing series of submodules

$$\langle 0 \rangle = M_n \subsetneq \cdots \subsetneq M_0 = M,$$

in which all quotients M_i/M_{i+1} are *simple* R -modules (exercise 3.5.4). The length of the series is the number of strict inclusions, and the composition factors are the quotients M_i/M_{i+1} .

Prove a Jordan-Hölder theorem for modules: any two finite composition series of a module have the same length and the same multiset of composition factors.

We say M has *length* m if M admits a finite composition series of length m . This notion is well-defined as a consequence of this proof.

Solution. a ■

Exercise 6.1.17. Prove that a k -vector space V has finite length as a module over k if and only if it is finite dimensional and that in this case its length is $\dim_k V$.

Solution. ifk ■

Exercise 6.1.18. Let M be an R -module of length m .

- Prove that every submodule N of M has finite length $n \leq m$.
- Prove that the descending chain condition for submodules holds in M (induct on the length).
- Prove that if R is an integral domain that is not a field and F is a free R -module, then F has finite length if and only if it is the 0-module.

Solution. wtf ■

Exercise 6.1.19. Let k be a field, and let $f(x) \in k[x]$. Prove that there exists a multiple of $f(x)$ in which all exponents of nonzero monomial additive factors are prime integers. (Hint: $k[x]/(f(x))$ is a finite dimensional vector space.)

Solution. a ■

Exercise 6.1.20. Let A, B be sets. Prove that the free groups $F(A), F(B)$ are isomorphic if and only if there exists a bijection $A \cong B$. (Hint: show that $F(A) \cong F(B) \implies F^{ab}(A) \cong F^{ab}(B)$.)

| **Solution.** a

